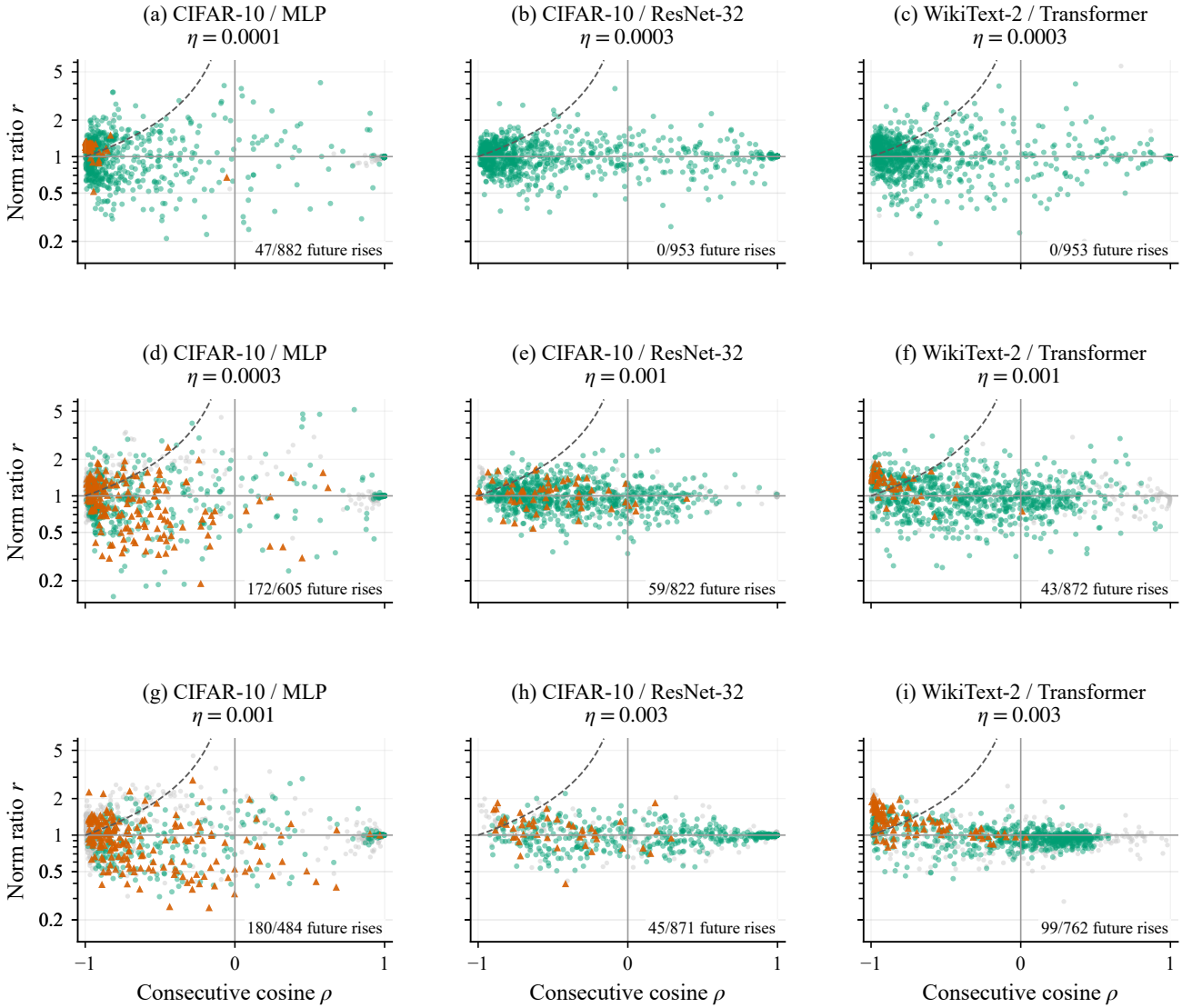

(a) CIFAR-10 / MLP
$\eta = 0.0001$
47/882 future rises
(b) CIFAR-10 / ResNet-32
$\eta = 0.0003$
0/953 future rises
(c) WikiText-2 / Transformer
$\eta = 0.0003$
0/953 future rises
(d) CIFAR-10 / MLP
$\eta = 0.0003$
172/605 future rises
(e) CIFAR-10 / ResNet-32
$\eta = 0.001$
59/822 future rises
(f) WikiText-2 / Transformer
$\eta = 0.001$
43/872 future rises
(g) CIFAR-10 / MLP
$\eta = 0.001$
180/484 future rises
(h) CIFAR-10 / ResNet-32
$\eta = 0.003$
45/871 future rises
(i) WikiText-2 / Transformer
$\eta = 0.003$
99/762 future rises
Norm ratio r
5
2
1
0.5
0.2
−1
0
1
Consecutive cosine ρ
Other states
Quiet: no future rise
Quiet: future rise
GD quadratic reference